INTERNSHIP REPORT

Title of the Report: AirBnb Clone - Backend API

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Bachelor of Technology (B.Tech)

Information Technology

Name of the Organization: Institute Internship Project

Internship Duration: January 2026 – May 2026

Name & Designation of Faculty Mentor: Dr. Sanjit Ningthoujam

Indian Institute of Information Technology Una

Academic Year: 2025–2026

Declaration

I hereby declare that the internship report entitled **"AirBnb Clone - Backend API"** submitted in partial fulfillment of academic requirements is an original work carried out by me during my institute internship project under the guidance of **Dr. Sanjit Ningthoujam**.

The work presented in this report has not been submitted previously for any degree, diploma, or certification in any institution.

Submitted By

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Acknowledgement

I would like to express my sincere gratitude to **Indian Institute of Information Technology Una** for providing me the opportunity to undertake this institute internship project.

I am deeply thankful to my faculty mentor, **Dr. Sanjit Ningthoujam**, for his continuous guidance, encouragement, and valuable suggestions throughout the duration of the project.

I would also like to thank all the faculty members of the Information Technology department for their support and motivation.

This project provided me with practical exposure to backend development, REST API design, authentication systems, database management, and enterprise application architecture using Spring Boot.

Finally, I would like to thank my friends and peers who directly or indirectly contributed to the successful completion of this project.

Abstract

This report presents the work carried out during the institute internship project from January 2026 to May 2026 at **Indian Institute of Information Technology Una**. The project focused on the development of an **AirBnb Clone Backend API** using Java and Spring Boot.

The project was designed to simulate the backend architecture of a real-world hotel booking platform similar to Airbnb. It included functionalities such as user authentication, hotel and room management, booking workflows, inventory tracking, and secure API communication using JSON Web Tokens (JWT).

The backend application followed a layered architecture using controllers, services, repositories, DTOs, and entities to ensure clean code organization and maintainability. PostgreSQL was used as the database system, while Spring Security and JWT were integrated to provide secure authentication and authorization.

The project provided practical exposure to enterprise backend development concepts, REST API design, database management, Spring ecosystem tools, and software engineering best practices.

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1. Introduction

1.1 Background of the Project

The hospitality and hotel booking industry has rapidly adopted digital platforms for room reservations, inventory management, and customer interaction. Modern booking applications require scalable backend systems capable of handling authentication, bookings, payments, and hotel management efficiently.

The AirBnb Clone Backend API project was developed to simulate the backend architecture of a production-level hotel booking platform. The project focuses on secure REST APIs, layered architecture, and efficient database handling using Spring Boot and PostgreSQL.

1.2 Objectives of the Internship

- To understand enterprise backend development workflows
- To develop a scalable backend API using Spring Boot
- To implement secure authentication using JWT
- To design RESTful APIs for hotel booking systems
- To gain practical exposure to database management and software engineering practices

1.3 Scope and Significance

The project demonstrates practical implementation of backend technologies used in real-world applications. The system can be extended further for frontend integration, payment gateways, cloud deployment, and microservices architecture.

2. Institute Profile

2.1 About the Institute

Indian Institute of Information Technology Una is an institute of national importance established to provide quality education and research opportunities in the field of Information Technology and Computer Science.

2.2 Vision and Mission

Vision: To become a globally recognized institution for excellence in technology education and research.

Mission: To provide high-quality technical education and encourage innovation, research, and professional development.

2.3 Department Overview

The Information Technology department focuses on software development, data structures, algorithms, databases, web technologies, artificial intelligence, and modern software engineering practices.

3. Internship Details

3.1 Duration and Type of Internship

The institute internship project was carried out from **January 2026 to May 2026** under the guidance of faculty mentor **Dr. Sanjit Ningthoujam**.

The internship focused on backend application development using modern Java technologies and enterprise software engineering concepts.

3.2 Roles and Responsibilities

- Designing backend architecture for the application
- Developing REST APIs using Spring Boot
- Implementing authentication and authorization using JWT
- Managing PostgreSQL database integration
- Creating DTOs and entity mappings
- Implementing booking and inventory workflows
- Handling exceptions and validations
- Testing API endpoints using Postman
- Managing project dependencies using Maven

3.3 Tools and Technologies Used

Technology	**Purpose**
Java 21	Core backend development
Spring Boot	REST API framework
Spring Security	Authentication and authorization
JWT	Secure API communication
Spring Data JPA	Database operations
PostgreSQL	Relational database management
ModelMapper	DTO and entity mapping
Lombok	Boilerplate code reduction
Maven	Dependency and build management
Postman	API testing
Git and GitHub	Version control and repository management

Table 3.1: Tools and Technologies Used

4. Work Carried Out / Project Description

4.1 Project Overview

The AirBnb Clone Backend API is a backend system designed for a hotel booking application similar to Airbnb. The system supports user authentication, hotel management, room management, booking operations, and inventory tracking.

The application follows a layered architecture consisting of controllers, services, repositories, DTOs, and entities. This approach improves maintainability, scalability, and clean separation of concerns.

4.2 Key Features

- User Signup and Login using JWT Authentication
- Hotel and Room Management APIs
- Booking Management System
- Inventory Tracking System
- Exception Handling using RestControllerAdvice
- DTO Mapping using ModelMapper
- Secure REST APIs using Spring Security

4.3 Methodology Followed

The project followed a modular backend development approach. Initially, database entities and relationships were designed for users, hotels, rooms, and bookings.

REST APIs were then developed using Spring Boot controllers and services. JWT authentication was integrated to secure application endpoints. PostgreSQL was used for persistent data storage, while Spring Data JPA handled database interactions.

DTOs were used to separate API request and response structures from database entities. Global exception handling mechanisms were implemented to improve API reliability and maintainability.

4.4 System Architecture

Client/User Interface
↓
REST API Controllers
↓
Service Layer (Business Logic)
↓
Repository Layer (JPA Repositories)
↓
PostgreSQL Database

Figure 4.1: Layered Architecture of the AirBnb Clone Backend API

4.5 Authentication Workflow

User Login Request
↓
Spring Security Authentication
↓
JWT Token Generation
↓
Token Sent to Client
↓
Authenticated API Access

Figure 4.2: JWT Authentication Workflow

4.6 Major Modules

- User Authentication Module
- Hotel Management Module
- Room Management Module
- Booking Management Module
- Inventory Management Module
- Security and Authorization Module

4.7 Results Achieved

- Successfully developed a production-style backend API
- Implemented secure JWT-based authentication
- Established CRUD operations for hotels and rooms

- Implemented booking workflows and inventory tracking
- Applied layered architecture for clean code organization
- Improved understanding of enterprise backend systems

5. Learning Outcomes

5.1 Technical Skills Acquired

- Java Backend Development
- Spring Boot Framework
- REST API Development
- Spring Security
- JWT Authentication
- PostgreSQL Database Management
- Spring Data JPA
- DTO and Entity Mapping
- Exception Handling
- Maven Dependency Management
- Git and GitHub Version Control

5.2 Soft Skills Developed

- Problem Solving
- Time Management
- Technical Documentation
- Debugging and Testing
- Software Design Thinking

5.3 Practical Exposure Gained

The internship project provided practical exposure to enterprise-level backend development workflows and modern software engineering practices used in scalable applications.

6. Challenges and Solutions

6.1 Problems Encountered

- Managing authentication securely using JWT
- Designing relationships between database entities
- Handling booking and inventory consistency
- Structuring scalable backend architecture
- Managing exception handling across APIs

6.2 Strategies Used to Overcome Them

- Used Spring Security for secure authentication workflows
- Applied layered architecture for better maintainability
- Used DTOs for clean request-response handling
- Implemented centralized exception handling
- Performed continuous API testing using Postman

7. Conclusion

The AirBnb Clone Backend API project provided valuable practical experience in backend software development using Java and Spring Boot.

The project enhanced understanding of REST APIs, authentication systems, database management, layered architecture, and enterprise software development workflows.

The internship contributed significantly to improving technical knowledge, debugging ability, software design skills, and professional development.

8. Suggestions and Recommendations

8.1 Future Improvements

- Integrate payment gateway services
- Add frontend application integration
- Deploy application on cloud platforms
- Implement caching for improved performance
- Extend the system using microservices architecture

8.2 Recommendations for Future Students

- Learn Java and OOP concepts thoroughly
- Understand REST API fundamentals
- Practice database design and SQL queries
- Gain familiarity with Spring Boot ecosystem
- Learn Git and version control practices